

Davide Bortoletto

GIS Analyst / GIS Specialist

Padua, Italy

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PROFESSIONAL PROFILE

GIS Analyst with a background in urban and regional planning (Master's Degree, LM-48, IUAV University of Venice) and professional experience in geospatial analysis, historical cartography and the integration of georeferenced data. I worked on defining the buffer zones of a UNESCO World Heritage site for a public client — from sourcing records in State Archives to georeferencing and diachronic landscape analysis — and on post-construction landscape monitoring of major infrastructure. Skilled in multi-criteria decision analysis (MCDA), remote sensing, LiDAR processing and process automation in Python.

PROFESSIONAL EXPERIENCE

GIS Analyst & CAD Draftsman | TERRE S.r.l.

Mestre (Venice), Italy · 02/2025 – 03/2026 (freelance / VAT collaboration)

Buffer zones of the UNESCO site "City of Vicenza and the Palladian Villas of the Veneto" — Municipality of Vicenza

- Geospatial analysis in QGIS to define the protection buffers, within a four-person working group: 24 villas analysed and 48 dossiers delivered.
- Sourcing of records at the State Archives of Vicenza, Venice and Verona, through direct liaison and on-site consultation of the cadastral registers, as well as at land-reclamation boards and municipal archives.
- Georeferencing of over 70 historical maps, from the 16th to the 20th century: iconography, Napoleonic and Austrian cadastres, Von Zach maps, IGM sheets, GAI aerial survey.
- Vectorisation of landscape features as points, lines and polygons, and diachronic analysis across successive time slices up to the present day, to measure changes in rural features, the road network and the hydrographic system.
- Integration of Veneto Geoportal datasets and of the municipal plan (PAT) layers — constraints, invariants, fragility and transformability; production of the map plates with QGIS print layouts and image post-processing (Photoshop, Lightroom, GIMP).
- Development of Python scripts to generate and populate the attribute tables of each villa's layers (identifier, administrative context, landscape feature, source, notes), filling the recurring fields across all selected layers in a single operation instead of entering them by hand.

Pedemontana Veneta motorway — post-construction landscape monitoring

- Updating of the landscape-impact assessment sheets section by section, with on-site photographic survey of the completed works and drafting of the assessment of their impact on the surrounding landscape.
- Field verification of the planting carried out along the route, counting the specimens and cross-checking them against the figures declared in the works progress reports.

High-speed railway — landscaping works

- 2D AutoCAD design of the planted areas of several sections, based on the external references (xrefs) supplied by the commissioning bodies and on planting schedules.

Intern | Studio Negri & Fauro Architetti Associati

Padua, Italy · 06/2022 – 07/2022

- CAD technical drawing of plans, sections and elevations of residential buildings (1:100), including energy-efficiency refurbishment works, and preparation of site plans (1:200 – 1:2000); site visits to the project areas and consultation of the municipal Interventions Plan and of the constraints, fragility and transformability maps (Valbrenta, Abano Terme, Noventa di Piave).
- On-site architectural surveys of the buildings concerned.

Intern | Municipality of San Giorgio delle Pertiche

Padua, Italy · 07/2019 – 08/2019

- Cadastral searches for taxation purposes through the municipal Territorial Information System (SIT).
- Checks that local businesses were up to date with their social-security contributions, via institutional portals (INAIL, Telemaco); creation and maintenance of a structured registry and dispatch of certified email (PEC).

Intern | Studio Tecnico Associato Battiston e Facco

Padua, Italy · 06/2017 – 08/2017

- Building surveys and CAD drafting of the resulting drawings following site inspections.

SELECTED GIS PROJECTS

Personal projects developed between April and July 2026 and published on [dav9817.github.io](https://github.com/dav9817)

Utility-scale solar PV siting — Province of Verona

Multi-criteria decision analysis (MCDA) to delineate the macro-areas most suitable for utility-scale photovoltaic development, combining land-use suitability, DEM-derived morphology and boolean exclusion masks (Natura 2000 network, water bodies, dense urban areas). The output is a Solar Hotspot map condensed into three districts.

Vegetation risk on a high-voltage power line — Vallegghi (Trento)

From LiDAR data, a geometric model of tree-fall risk on the conductors of a new overhead HV line is built; three management strategies are compared by measuring residual risk over five years, showing the superiority of the predictive geometric approach.

NO2 and night-time lights in the Po Basin

A reproducible pipeline built on Sentinel-5P/TROPOMI and Suomi-NPP/VIIRS data: cloud extraction on Google Earth Engine, statistical analysis in Python, cartography in QGIS. Night-time lights explain about 41% of the spatial variability of NO2; the remainder is driven by the shape of the Po Valley basin.

Settlement suitability on Mars — MCDA on NASA/USGS data

Multi-criteria analysis in QGIS across five criteria reclassified on a common 0–5 scale (terrain morphology, access to subsurface water ice, surface geology, insolation, mean temperature), from an engineering rather than a biological perspective. The model yields a composite suitability map, published as an interactive 3D globe.

TECHNICAL SKILLS

- **GIS & Geospatial Analysis:** QGIS, multi-criteria analysis (MCDA, weighted overlay), geoprocessing, zonal statistics, raster reclassification, geodatabase management, print layouts and cartographic production; georeferencing of historical cartography, digitising and vectorisation, multi-temporal analysis; integration of georeferenced data (Veneto Geoportal, municipal SIT, NASA/USGS open data); basic knowledge of ArcGIS (academic use).
- **Remote Sensing:** Google Earth Engine, basic use on Copernicus/Sentinel-5P (TROPOMI) and Suomi-NPP/VIIRS data; eCognition (OBIA) and ERMapper (academic use).
- **LiDAR & Terrain Models:** point-cloud processing in QGIS, DEM/DSM, morphological analysis (slope, aspect) — in portfolio projects.
- **Historical and archival sources:** iconography, Napoleonic and Austrian cadastres and their registers, Von Zach maps, IGM sheets, GAI aerial survey, MAPIRE.
- **Programming & Automation:** Python — scripting for GIS process automation and statistical analysis.
- **Spatial Databases:** PostgreSQL/PostGIS and spatial SQL (in training).
- **Web Mapping & Publishing:** interactive web maps, vector tiles, HTML/CSS/JavaScript — in portfolio projects.
- **CAD & Technical Drafting:** Autodesk AutoCAD, ArchiCAD, 2D technical drawing. **Graphics:** Adobe Photoshop, Adobe Lightroom, GIMP, Pixelmator Pro. **Systems:** Microsoft Office; Windows, macOS.

EDUCATION

Master's Degree in Planning and Policies for the City, the Environment and the Landscape (LM-48) — IUAV University of Venice

Venice, Italy · 2020 – 2024 · Final mark: 106/110

Track: City-Territory-Landscape. Training in GIScience, territorial information systems and remote sensing, territorial and landscape planning, land governance and heritage protection.

Bachelor's Degree in Land Reorganisation and Landscape Protection (L-21) — University of Padua

Padua, Italy · 2017 – 2020 · Final mark: 103/110

Track: Land Protection and Reorganisation (TRT). Training in GIS, CAD, territorial representation, hydrogeological risk and territorial planning.

Surveyor's Diploma (Geometra) — Technical Institute, Construction, Environment and Territory

I.T.I.S. "G. Belzoni", Padua, Italy · 2013 – 2017 · Final mark: 80/100

CERTIFICATIONS, LICENCES AND LANGUAGES

- Remote pilot of unmanned aircraft (UAS) — "open" category, subcategories A1/A3 (EASA/ENAC, EU Reg. 2019/947). Certificate no. ITA-RP-000081170acx, valid until 13/07/2031.
- Driving licence — categories A and B.
- Languages: Italian native speaker; English B1 level.